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Total Number of Pages: 2

B.Tech.

MA101

MID TERM EXAMINATION-2025

Mathematics-I

Branch: ALL

Semester- 1st

Time: 1 Hour

Max marks: 20

Answer all Questions from Part – A and Part-B

The figures in the right-hand margin indicate marks.

PART-A

Q1. Answer the following questions:

[5x2=10]

a) Is Rolle's theorem applicable to the function $f(x) = |x|$ in $[-1, 1]$? Justify your answer. [CO1, L3]

b) Find the value of c of the mean value theorem if $f(x) = 2x^2 + 3x + 4$ in $[1, 2]$. $c = 1, -\frac{5}{2}$ [CO1, L2]

c) If $u = \log \left\{ \frac{x^4 + y^4}{x + y} \right\}$ using Euler's theorem show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$ [CO2, L2]

d) Show that the following limit does not exist $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$. [CO2, L2]

$$u = 1 + 25 + 20h$$

$$u = (1 + 25 + 20h) + (1 + 25 + 20h) = 2 + 20h$$

e) Find the stationary points of the function

$$f(x, y) = 2(x^2 - y^2) - x^4 + y^4$$

[CO2, L2]

PART - B

Q2. Answer any one of the following questions:

[5]

a) Find all the asymptotes of the following curve

$$f(x) = x^3 + 4x^2y + 4xy^2 + 5x^2 + 15xy + 10y^2 - 2y + 1 = 0$$

(3)

(3)

(3)

(2)

(2)

(2)

(1)

(0)

[CO1, L2]

(b) Show that the radius of curvature for a cycloid

$$x = a(t + \sin t), y = a(1 - \cos t) \text{ is } \rho = 4a \cos\left(\frac{t}{2}\right)$$

[CO1, L2]

Q3. Answer any one of the following questions:

[5]

a) Expand the function $f(x, y) = x^2 + xy - y^2$ by Taylor's theorem in powers of $(x - 1)$ and $(y + 2)$.

[CO2, L2]

b) Discuss the relative maximum and relative minimum of

$$f(x, y) = xy + \frac{9}{x} + \frac{3}{y}$$

[CO2, L2]

$4x^2 + 5x$

$$-5 + 5(y+2) + (x-1)^2 - (y+2)^2 + (x-1)(y+2)$$

Total number of printed pages – 3

1ST SEMESTER REGULAR EXAMINATION – 2025

SUBJECT NAME: MATHEMATICS-I

BRANCH: B.TECH. ALL BRANCHES

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Discuss applicability of Rolle's Theorem to the function $f(x) = x $ in $[-1,1]$.	CO1	2	2
(b)	State Lagrange's mean value theorem	CO1	2	1
(c)	Find the asymptotes to the polar curve $r\theta = a$	CO1	2	2
(d)	State Euler's theorem for the homogenous function of two variables.	CO2	2	1
(e)	Evaluate the value of $\beta(4,2)$.	CO2	2	2
(f)	Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y}$ does not exist.	CO2	2	2
(g)	Find $\frac{d\vec{r}}{dt}$ at $t = 1$, where $\vec{r} = xy^2\hat{i} + yz^2\hat{j} + zx^2\hat{k}$ and $x = 1, y = t, z = t$.	CO3	2	2
(h)	Find unit normal vector at $(1,2,-1)$ of the surface $x^3 + y^3 + 3xyz = 3$.	CO3	2	3
(i)	Find the directional derivative of the function $\phi(x,y,z) = x^2 + y^2 + z^2$ at the point $(1,1,1)$ in the direction $\hat{i} + 2\hat{j} + 2\hat{k}$.	CO3	2	2
(j)	Prove that $\vec{\nabla}(uv) = u \vec{\nabla}(v) + v \vec{\nabla}(u)$, where u and v are scalar functions	CO4	2	2
(k)	Calculate $\oint (x dy - y dx)$ around the circle $x^2 + y^2 = 1$	CO4	2	2
(l)	State Gauss's divergence theorem.	CO4	2	1

(m)	Prove that $f(x) = -\sin(x^2)$ is an even function.	CO5	2	2
(n)	Find the Fourier integral of $f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$	CO5	2	2
(o)	Prove that the period of the function $f(x) = \sin x$ is 2π .	CO5	2	2

Part-B		CO	Marks	BL
Q2.	Answer any two:			
(a)	If $f(x) = \sin x$ and $g(x) = \cos x$ in $[a, b]$, then show by Cauchy's mean value theorem that the value of $c \in (a, b)$ is the arithmetic mean of a and b .	CO1	7	3
(b)	Find all types of asymptotes to the curve $x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$	CO1	7	3
(c)	Find the radius of curvature of the curve given by $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$.	CO1	7	2
Q3.	Answer any two:			
(a)	Show that the function $f(x, y) = \begin{cases} \frac{xy}{x^2 + 2y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$ is not continuous at $(0, 0)$, but its partial derivatives f_x and f_y exist at $(0, 0)$.	CO2	7	2
(b)	Find the Taylor's series expansion of the function $f(x, y) = 2x^3 + 3y^3 - 4x^2y$ about the point $(1, 2)$.	CO2	7	2
(c)	If $p > -1, q > -1$, prove that $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$.	CO2	7	3
Q4.	Answer any two:			
(a)	Calculate the line integral $\int_C \vec{f} \cdot d\vec{r}$ where $\vec{f} = [y^2, -x^2]$ and C is the straight line from $(0, 0)$ to $(1, 4)$.	CO3	7	2
(b)	Prove that the vector function $\vec{f} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is irrotational and solenoidal.	CO3	7	2
(c)	Show that the integral $\int_C \vec{f} \cdot d\vec{r} = \int_C (3x^2 dx + 2yz dy + y^2 dz)$ is independent of the path and then evaluate the integral value when the initial point and terminal point of C are $A(0, 1, 2)$ and $B(1, -1, 7)$.	CO3	7	3
Q5.	Answer the following:			
(a)	Using Green's theorem evaluate $\oint_C (P dx + Q dy)$, where $P = x \cosh 2y$, $Q = 2x \sinh 2y$.	CO4	7	3

	$Q = x^2 \sinh 2y$ and C is the boundary of the region $x^2 \leq y \leq x$.			
(b)	Using Gauss's divergence theorem evaluate the surface integral $\iint_S \vec{f} \cdot \hat{n} dS$, where $\vec{f} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ and S is the surface of the rectangular parallelepiped bounded by $x = 0, x = a, y = 0, y = b, z = 0, z = c$. 2m	CO4	7	3.
Q6.	Answer the following:			
(a)	Find the Fourier series of the functions $f(x) = \begin{cases} k & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$ and hence prove that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$	CO5	7	3
(b)	Find the Fourier transform of the function $f(x) = \begin{cases} e^x, & \text{if } x < 0 \\ e^{-x}, & \text{if } x > 0. \end{cases}$	CO5	7	2

$$\begin{matrix} \vec{f} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \hline f_1 & f_2 & f_3 \end{matrix}$$

$$\begin{aligned} \vec{e} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) - \vec{j} \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} \right) + \vec{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \\ \vec{e} (3x - 3x) - \vec{j} (3y - (-1 + 3y)) + \vec{k} (3y - 2z + 2z - 3y) \end{aligned}$$

$$\begin{aligned} \frac{\partial f_3}{\partial x} &= 3y - 2z \\ \frac{\partial f_1}{\partial z} &= -2z + 3y \\ \frac{\partial f_2}{\partial x} &= 3z + 2y - 2x - 3x \end{aligned}$$

$$\begin{aligned} x \cdot \cosh 2x \\ x \cdot \sinh 2x \cdot \frac{1}{4x} \\ 1 \cdot \cosh 2x \cdot \frac{1}{16x} \\ + x \cdot \sinh 2x \cdot \frac{1}{2} \\ - 1 \cdot \cosh 2x \cdot \frac{1}{4} \end{aligned}$$

$$\frac{1}{2} \times \frac{1}{24} \times \frac{\pi}{4}$$